

Differential Geometry

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Contents

Chapter 1

Euclidean Spaces

1.1 Smooth Functions on Euclidean Space

Note that coordinates on R^n are notated as $x^1, \dots, x^n$ so $p = (p^1, \dots, p^n)$ is a point in some open set U in R^n .

1.1.1 Smooth vs Analytic function

Definition 1.1 (Smooth functions). $f : U \rightarrow \mathbb{R}$ is C^k at $p \in U$ if it's partial derivatives,

$$\frac{\partial^j f}{\partial x^{i_1} \dots \partial x^{i_j}}$$

of all orders $j \leq k$ exists and are continuous at p . Similarly f is said to be C^∞ or smooth if this is true for all $k \geq 0$.

Note. The i 's in the denominator allow repeated index.

Example. 1. A C^0 function on U is any continuous function on U .

2. If $f : \mathbb{R} \rightarrow \mathbb{R}$ be $f(x) = x^{1/3}$ then f is C^0 but not C^1 at $x = 0$.

3. If we take the integral of $x^{1/3}$ that function is C^1 but not C^2 at $x = 0$.

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Definition 1.2 (Real-analytic function). A function f is real-analytic at p if in some neighborhood of p it is equal to it's Taylor series at p . So we have,

$$f(x) = f(p) + \sum_i \partial_{x^i} f(x^i - p^i) + \frac{1}{2!} \sum_{i,j} \partial_{x^i} \partial_{x^j} f(p)(x^i - p^i)(x^j - p^j) + \dots$$

Note. Here, the function is not just approximated but is equal to it's Taylor series at that neighborhood.

Remark. A real-analytic function is **necessarily** smooth. We see this as we can infinitely differentiate the power series and because the function is equal to it we have infinitely differentiability of the function. Note however that a smooth function **need not** be real-analytic.

Example. Here is a smooth function that is not real-analytic,

$$f(x) = \begin{cases} e^{-1/x} & \text{for } x > 0 \\ 0 & \text{for } x \leq 0 \end{cases}$$

We can easily show by induction this is smooth. But note the Taylor series at $x = 0$ in any neighborhood of zero is identically zero since all the derivatives $f^{(k)}(0)$ equal to 0. But if we take any neighborhood around the origin for any $x > 0$ note that f is a non-zero value. Hence, the function can never be equal to the Taylor series at any neighborhood around the origin. $\diamond$

1.1.2 Taylor's Theorem with Remainder

A subset S of R^n is star-shaped with respect to a point p in S if for every $x \in S$, the line segment from p to x lies in S .

Lemma 1.3 (Taylor's theorem with remainder). Let f be a smooth function on an open subset U of R^n star-shaped wrt a point $p = (p^1, \dots, p^n)$. Then there are functions $g_1(x), \dots, g_n(x)$ smooth, such that,

$$f(x) = f(p) + \sum_{i=1}^n (x^i - p^i) g_i(x), \quad g_i(p) = \partial_{x^i} f(p)$$

Remark. The usual Taylor's theorem linear approximation gives us a remainder term, but here the "error" correction is absorbed using the functions g_i defined using x^i th derivative of f .

Proof. As U is star shaped for any x in U there is a line segment connecting x and p , i.e $p + t(x - p)$ for $t \in [0, 1]$. So $f(p + t(x - p))$ is defined. Now we have,

$$\frac{d}{dt} f(p + t(x - p)) = \sum (x^i - p^i) \frac{\partial f}{\partial x^i}(p + t(x - p))$$

Now integrating both sides we get,

$$f(p + t(x - p)) \Big|_0^1 = \sum (x^i - p^i) \int_0^1 \partial_{x^i}(p + t(x - p)) dt$$

Now take,

$$g_i(x) = \int_0^1 \partial_{x^i}(p + t(x - p)) dt$$

Now g_i is smooth and we have,

$$f(x) - f(p) = \sum_x (x^i - p^i) g_i(x)$$

and,

$$g_i(p) = \partial_{x^i} f(p)$$

Now when $n = 1$ and $p = 0$ the lemma gives us,

$$f(x) = f(0) + x g_1(x)$$

and as g is smooth we can write,

$$g_i(x) = g_i(0) + x g_{i+1}(x)$$

So doing this repeatedly we get,

$$\begin{aligned}
 f(x) &= f(0) + x(g_1(0) + xg_2(x)) \\
 &= f(0) + xg_1(0) + x^2(g_2(0) + xg_3(x)) \\
 &\quad \vdots \\
 &= f(0) + g_1(0)x + g_2(0)x^2 + \cdots + g_i(0)x^i + g_{i+1}(x)x^{i+1}
 \end{aligned}$$

□

1.1.3 Problems

Problem. Consider $f(x) = e^{-1/x}1_{[0,\infty)}$, note that this function is smooth and we have $f^{(k)}(0) = 0$ for all $k \geq 0$. So the function is very flat at 0.

Problem. A diffeomorphism of an open interval with $\mathbb{R}$. A smooth function $f : U \rightarrow V$ is called a diffeomorphism if it is bijective and has a smooth inverse.

(a). Show that $f : [-\pi/2, \pi/2] \rightarrow \mathbb{R}, f(x) = \tan x$ is a diffeomorphism. First we need to show it's bijective. For injectivity note we have $f(x) = f(y) \Rightarrow \tan(x) = \tan(y)$ but as our domain is $[-\pi/2, \pi/2]$ and we know that $x = \tan y$ implies that $y = 2\pi i + x$ but the only value of i for which y is within our domain is $i = 0$ which gives us $x = y$.

Now note that the codomain of the tan function is $\mathbb{R}$. So there exist some x such that $\tan(x) = y$. Now as the cycle is length 2π there exists some $x_0 \in [-\pi/2, \pi/2]$ such that $\tan(x_0) = \tan(x)$. So we have $(x_0) = y$, so we have a surjection.

Now we need to show the inverse is a smooth map. Note the inverse of $\tan(x)$ is $\tan^{-1}(x)$ whose derivative is $\frac{1}{1+x^2}$. Note that however $\frac{1}{1+x^2}$ is a smooth map. Hence, $\tan^{-1}$ is a smooth function.

(b). We need a function $h : [a, b] \rightarrow [-1, 1]$ that is a diffeomorphism. Just consider the linear function,

$$y = 1 + \frac{2(x-b)}{(b-a)}$$

Problem. $f : \mathbb{R} \rightarrow \mathbb{R}$ and $f(x) = x^3$. Then we have f is bijective smooth map, but $f^{-1}(x) = x^{1/3}$ is not a smooth inverse map. So we can have a bijective smooth map whose inverse is not smooth.

Remark. For complex functions not that a bijective holomorphic (complex-differentiable) map necessarily has a holomorphic inverse.

1.2 Tangent Vectors in $\mathbb{R}^n$ as Derivations

We traditionally define a vector,

$$v = \begin{bmatrix} v^1 \\ v^2 \\ v^3 \end{bmatrix}$$

as an arrow from the origin pointing to the point.

A secant plane in a surface in R^3 is a plane determined by three points of the surface. As the three points approach a point p the limit of the plane is the tangent plane. So a vector at p is tangent to a surface if it lies in the tangent plane at p .